Reuse Gate: pilot and continuation plan

Team ZtotheZ · Since AI Hackathon 2026, Bayer challenge "Build Less, Build Better" · 8 November 2026

Proposal. An 8-week pilot with one analytics team in one Bayer division, on real BI metadata, to measure whether Reuse Gate avoids new builds without slowing requesters down. A go or no-go decision at week 8 on thresholds agreed before the start.

1. Where we are

The prototype runs on Red Hat OpenShift AI with Bayer's synthetic Builder Data Pack: 360 reports, 56 KPI definitions, 75 semantic models, 150 business requests. On that data, 57 of 150 requests avoid a new build and all 1,098 cited rows exist. These are modelled results on synthetic data. The pilot replaces them with measured ones.

Built and tested: the decision pipeline with cited evidence and seven release gates, the clarification loop, human approval with owner and review date, least privilege for seven roles enforced by the API, the analyst dashboard studio, measured configuration changes, the executive report, the operator wall, email through n8n, a KServe decision service and a gated Tekton pipeline.

2. Scope

In scope	Out of scope
One division and one analytics team (suggested: a commercial analytics team with an active Power BI estate) New report requests for that team for 8 weeks Read-only metadata: workspaces, reports, semantic models, measures, owners, usage	Changing or deleting any report automatically Business data values (only metadata and usage counts) Other divisions, until the go decision

3. Plan

Weeks	Work	Exit criterion
0 to 1	Agree thresholds and owners. Security review. Connect single sign-on (only the identity function changes).	Signed pilot charter
1 to 3	Replace the data pack loader with the Power BI or Fabric metadata scanner and usage logs. Map the division's KPI catalogue and owners. Re-run the evaluation on real metadata.	Citation validity 1.0 on real metadata

3 to 4	Shadow mode: requests go through the normal process, and Reuse Gate's recommendation is recorded beside it without being shown.	Agreement with what the team actually did
4 to 8	Live mode: requesters see the answer card, approvers decide in Reuse Gate, developers build from the spec.	Metrics below, weekly
8	Review with the team, the division's BI lead and governance. Go or no-go.	Decision on record

4. What we measure

Metric	How	Suggested go threshold
New builds avoided	Approved reuse, repair, extend or migrate decisions / all decided requests	at least 20%
Recommendation accepted	Approver agrees with the recommended action	at least 70%
Time to first answer	Request filed to answer card	under 1 working day, against today's baseline
Owner coverage	New and touched reports with an active owner and review date	100% of approved items
Citation validity	Cited rows that exist in the metadata	1.0
Requester satisfaction	Two-question survey after each answer	at least 4 of 5

A 20-row human gold set is already prepared in the repository; the pilot team labels real requests the same way, so the recommendation quality is measured, not only the rule compliance.

5. Roles

- **Business requesters** file requests and answer questions.
- **Analysts** check evidence and build previews in the dashboard studio.
- **Approver** (the team's BI lead) decides, with owner and review date.
- **Developers** build from approved specs and propose rule changes, measured before review.
- **Governance and IT** own access, the decision log and the platform.

6. Platform, security and governance

- Runs in the division's Red Hat OpenShift AI project: the KServe decision service, the API and the UI, with the existing Tekton gates (tests, governance, evaluation, security) before every rollout.
- Single sign-on replaces access codes; roles come from directory groups. Least privilege stays enforced by the API.

- Language models only read requests and word text. In the pilot they run on an approved endpoint (or in-cluster on a GPU node); with no model, decisions are unchanged.
- Read-only metadata access. No business data values leave the source systems.
- Before scale-up: approval rights by domain, separation of duties on requests, and a tamper-evident central decision log.

7. Risks

Risk	Mitigation
Metadata is messier than the data pack	Shadow mode first; the gates name what is missing instead of guessing
Requesters bypass the gate	Make the answer faster than building; the BI lead routes new requests through it
Rules too strict or too loose	Developers propose changes, measured on all requests, approved by someone else
Model provider unavailable	Tiered fallback to rules; decisions do not depend on a model

8. After a go decision

Extend to the division's other analytics teams in quarter one, with .pbip export from the dashboard studio and a link from the BI request portal. Each new division plugs in through its own catalogue file (KPI vocabulary, products, markets), with no change to the decision logic.